

Asynchronous Buck — MPPT Power Stage

Design Calculations

Givens

$$\begin{array}{llll}
 V_{in} = 18 \text{ V} & V_{oc} = 22 \text{ V} & V_{out} = 12 \text{ V} & I_{out} = 1.5 \text{ A} \\
 L = 560 \text{ }\mu\text{H} & C_{in} = 220 \text{ }\mu\text{F} & C_{out} = 1000 \text{ }\mu\text{F} & \text{ESR} = 0.1 \text{ }\Omega \\
 R_{ds(on)} = 0.117 \text{ }\Omega & Q_g = 70 \text{ nC} & V_F = 0.55 \text{ V} & \text{DCR} = 0.3 \text{ }\Omega \\
 R_1 = 1 \text{ k}\Omega & R_2 = 220 \text{ }\Omega & R_3 = 680 \text{ }\Omega & V_Z = 10.2 \text{ V}
 \end{array}$$

1. Switching frequency

$$f_{sw} = \frac{f_{APB}}{2^n} = \frac{80 \times 10^6}{2^{13}} = \frac{80 \times 10^6}{8192} = 9765.6 \text{ Hz}$$

$$\Delta D_{LSB} = \frac{1}{2^{13}} = 1.22 \times 10^{-4} = 0.0122 \% \quad \Delta V_{LSB} = V_{in} \Delta D_{LSB} = 18 \times 1.22 \times 10^{-4} = 2.2 \text{ mV}$$

2. Duty cycle

$$\begin{aligned}
 D (V_{in} - I_{out} R_{ds(on)} - V_{out}) &= (1 - D) (V_{out} + V_F) \\
 D &= \frac{V_{out} + V_F}{V_{in} - I_{out} R_{ds(on)} + V_F} = \frac{12 + 0.55}{18 - (1.5)(0.117) + 0.55} = \frac{12.55}{18.374} = 0.683 \\
 D_{ideal} = \frac{V_{out}}{V_{in}} &= \frac{12}{18} = 0.667 \quad t_{on} = \frac{D}{f_{sw}} = 69.9 \text{ }\mu\text{s} \quad T = \frac{1}{f_{sw}} = 102.4 \text{ }\mu\text{s}
 \end{aligned}$$

3. Inductor current

$$\begin{aligned}
 \Delta I_L &= \frac{V_{out}(1 - D)}{L f_{sw}} = \frac{12(0.317)}{(560 \times 10^{-6})(9765.6)} = \frac{3.804}{5.4688} = 0.696 \text{ A} \\
 I_{pk} &= I_{out} + \frac{\Delta I_L}{2} = 1.5 + 0.348 = 1.85 \text{ A} \quad I_{min} = 1.5 - 0.348 = 1.15 \text{ A} \\
 I_{L(rms)} &= \sqrt{I_{out}^2 + \frac{\Delta I_L^2}{12}} = \sqrt{2.25 + \frac{0.4844}{12}} = \sqrt{2.2904} = 1.513 \text{ A} \\
 I_{out(crit)} &= \frac{\Delta I_L}{2} = 0.348 \text{ A} \quad R_{load(crit)} = \frac{V_{out}}{I_{out(crit)}} = \frac{12}{0.348} = 34.5 \text{ }\Omega \quad R_{load} = \frac{12}{1.5} = 8 \text{ }\Omega < 34.5 \text{ }\Omega
 \end{aligned}$$

4. Capacitors

$$\Delta V_C = \frac{\Delta I_L}{8f_{sw}C_{out}} = \frac{0.696}{8(9765.6)(10^{-3})} = 8.9 \text{ mV} \quad \Delta V_{ESR} = \Delta I_L \cdot \text{ESR} = 0.696(0.1) = 69.6 \text{ mV}$$

$$\Delta V_{out} = \Delta V_C + \Delta V_{ESR} = 78.5 \text{ mV}_{pp}$$

$$I_{Cin(rms)} = I_{out} \sqrt{D(1-D)} = 1.5 \sqrt{(0.683)(0.317)} = 1.5 \sqrt{0.2165} = 0.698 \text{ A}$$

$$\Delta V_{in} = \frac{I_{out}D(1-D)}{C_{in}f_{sw}} = \frac{1.5(0.2165)}{(220 \times 10^{-6})(9765.6)} = \frac{0.3248}{2.1484} = 0.151 \text{ V}$$

5. Gate drive

$$V_Z = 2 \times V_{Z(1N4733A)} = 2(5.1) = 10.2 \text{ V} < 20 \text{ V} = |V_{GS(max)}| \quad V_{GS(unclamped)} = -V_{oc} = -22 \text{ V}$$

$$I_{R3} = \frac{V_Z}{R_3} = \frac{10.2}{680} = 15.0 \text{ mA} \quad I_{R2} = \frac{V_{in} - V_Z - V_{ce(sat)}}{R_2} = \frac{18 - 10.2 - 0.2}{220} = 34.5 \text{ mA}$$

$$I_Z = I_{R2} - I_{R3} = 34.5 - 15.0 = 19.5 \text{ mA} \quad P_Z = V_Z I_Z = 10.2(19.5 \times 10^{-3}) = 0.199 \text{ W} < 2 \text{ W}$$

$$P_{R2} = \frac{(V_{in} - V_Z - V_{ce})^2}{R_2} D = \frac{7.6^2}{220} (0.683) = 0.179 \text{ W} \quad P_{R3} = \frac{V_Z^2}{R_3} D = \frac{10.2^2}{680} (0.683) = 0.105 \text{ W}$$

$$I_B = \frac{V_{GPIO} - V_{BE}}{R_1} = \frac{3.3 - 0.7}{1000} = 2.6 \text{ mA} \quad \beta_{req} = \frac{I_{R2}}{I_B} = \frac{34.5}{2.6} = 13.3 \ll h_{FE} = 250$$

$$t_{on} = \frac{Q_g}{I_Z} = \frac{70 \times 10^{-9}}{19.5 \times 10^{-3}} = 3.6 \text{ } \mu\text{s} \quad t_{off} = \frac{Q_g}{I_{R3}} = \frac{70 \times 10^{-9}}{15.0 \times 10^{-3}} = 4.7 \text{ } \mu\text{s}$$

$$\frac{t_{on} + t_{off}}{T} = \frac{8.3}{102.4} = 8.1 \% \quad (Q_g = 180 \text{ nC} : t_{on} = 9.2 \text{ } \mu\text{s}, t_{off} = 12.0 \text{ } \mu\text{s}, 20.7 \%)$$

6. Losses

$$P_{sw} = \frac{1}{2} V_{in} I_{out} (t_{on} + t_{off}) f_{sw} = \frac{1}{2} (18)(1.5)(8.3 \times 10^{-6})(9765.6) = 1.09 \text{ W}$$

$$P_{cond} = I_{out}^2 R_{ds(on)} D = (1.5)^2 (0.117)(0.683) = 0.180 \text{ W} \quad P_D = V_F I_{out} (1-D) = 0.55(1.5)(0.317) = 0.261 \text{ W}$$

$$P_L = I_{L(rms)}^2 \text{DCR} = (1.513)^2 (0.3) = 0.687 \text{ W} \quad P_{drive} = V_{in} I_{R2} D = 18(34.5 \times 10^{-3})(0.683) = 0.424 \text{ W}$$

$$P_{sh} = 2I_{out}^2 R_{sh} = 2(1.5)^2 (0.02) = 0.090 \text{ W}$$

Term	W
P_{sw}	1.09
P_{cond}	0.18
P_D	0.26
P_L	0.69
P_{drive}	0.42
P_{sh}	0.09
P_{loss}	2.73

$$P_{out} = V_{out} I_{out} = 18 \text{ W} \quad \eta = \frac{P_{out}}{P_{out} + P_{loss}} = \frac{18}{20.73} = 86.8 \%$$

$$Q_g = 180 \text{ nC} : P_{sw} = 2.73 \text{ W}, P_{loss} = 4.37 \text{ W}, \quad \eta = \frac{18}{22.37} = 80.5 \%$$

7. INA226

$$I_{max} = \frac{V_{sh(FS)}}{R_{sh}} = \frac{81.92 \times 10^{-3}}{0.1} = 0.819 \text{ A} < 1.5 \text{ A} \quad \frac{81.92 \times 10^{-3}}{0.02} = 4.10 \text{ A} > 1.5 \text{ A}$$

$$\text{Current_LSB} = \frac{I_{max,exp}}{2^{15}} = \frac{1.5}{32768} = 45.8 \mu\text{A} \rightarrow 50 \mu\text{A}$$

$$\text{CAL} = \frac{0.00512}{\text{Current_LSB} \cdot R_{sh}} = \frac{0.00512}{(50 \times 10^{-6})(0.02)} = \frac{0.00512}{10^{-6}} = 5120$$

$$\text{Power_LSB} = 25 \cdot \text{Current_LSB} = 1.25 \text{ mW} \quad V_{bus(LSB)} = 1.25 \text{ mV}, \quad V_{bus(FS)} = 40.96 \text{ V} > V_{oc}$$

8. Control loop

$$f_c = \frac{1}{2\pi\sqrt{LC_{out}}} = \frac{1}{2\pi\sqrt{(560 \times 10^{-6})(10^{-3})}} = \frac{1}{2\pi(7.483 \times 10^{-4})} = 212.7 \text{ Hz}$$

$$\frac{f_{sw}}{f_c} = \frac{9765.6}{212.7} = 45.9 \quad \tau = \frac{1}{2\pi f_c} = 0.75 \text{ ms} \quad 5\tau = 3.7 \text{ ms} \ll t_{P\&O} = 100 \text{ ms}$$

$$\Delta D_{step} = 15 \cdot \Delta D_{LSB} = 1.83 \times 10^{-3} \quad \Delta V_{step} = V_{in} \Delta D_{step} = 33 \text{ mV}$$

$$I_{inrush} = C_{out} \frac{dV}{dt} = 10^{-3} \left(\frac{12}{0.2} \right) = 60 \text{ mA} \ll I_{out}$$

$$D_{PWM} = 1 - D \quad (\text{inverting stage}) \quad D \in [0, 1] \quad (\text{no bootstrap constraint})$$

9. Thermal

$$P_{Q1} = P_{sw} + P_{cond} = 1.09 + 0.18 = 1.27 \text{ W}$$

$$\Delta T_{JA} = P_{Q1} R_{\theta JA} = 1.27(62) = 79^\circ\text{C} \quad \Delta T_{JA(hs)} = 1.27(20) = 25^\circ\text{C}$$

$$\Delta T_D = P_D R_{\theta JA} = 0.261(30) = 7.8^\circ\text{C} \quad \Delta T_L = P_L R_{\theta} = 0.687(30) = 21^\circ\text{C}$$